AgriSync: Predictive Timing System for Nutrient Optimization

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Problem Statement

Surging fertilizer costs and wasteful spreading methods are forcing farmers to use less nutrients, lowering crop yields and threatening global food security.

Problem Understanding

- **Target Population:** Smallholder and medium-scale agrarian farmers who rely on manual broadcasting or fixed-calendar fertilizing.
- **Core Drivers:** Climate volatility induces sudden unseasonal rain or severe dry spells, breaking conventional calendar assumptions.
- **Compounded Impact:** Applying before heavy rainfall washes away nutrients (eutrophication); applying on parched soil prevents root uptake, burning crops and gutting operational ROI.
- **Global Vulnerability:** Smallholder farmers face up to 30–40% input waste due to premature runoff or volatilization.
- **Economic Bottleneck:** Rising input prices compress profit margins, driving intentional under-application and yield loss.

Existing Gaps & Challenges

- **Static Calendar Reliance:** Legacy schedules fail to reflect real-time microclimate and soil moisture fluctuations.
- **High Cost of Smart Agri-Tech:** Existing precision solutions (ISOBUS rate controllers, drone NDVI) are economically out of reach for smallholders.
- **The Timing Blindspot:** Most existing tools advise *how much* to spread, but fail to predict *the exact optimal window* when roots can actually absorb it.

Our Proposed Solution

- **Dynamic Timing Engine:** An algorithmic recommendation system identifying the precise 24–48 hour application window for maximum nutrient uptake.
- **Dual-Factor Input Ingestion:** Continuously evaluates high-resolution hyper-local weather forecasts alongside real-time soil moisture dynamics.
- **Actionable Advisory:** Delivers immediate "Go / Hold / Adjust Dose" decisions through SMS and an accessible lightweight dashboard.

Technical Approach

Data Layer: Low-Cost Capacitive Soil Sensors + OpenWeather/IMD APIs

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Processing Engine: Flask/FastAPI Backend + ML Timing Regression / Rule-Based Model

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Delivery Layer: Flutter Mobile UI / Twilio SMS Gateway for Field Alerts

Tech Stack

Frontend: Flutter / React PWA

Backend: Python (FastAPI)

Database: PostgreSQL & TimescaleDB

AI/ML & IoT: Scikit-learn, ESP32, Open-Meteo API

Innovation & Uniqueness

- **Shift to Precision:** Pivots precision agriculture from expensive spatial machinery to time-based uptake optimization.
- **Microclimate & Root Hydration Coupling:** Eliminates leaching risk by requiring simultaneous moisture and low rain-risk criteria.
- **Low Barrier to Entry:** Delivers high-tier predictive intelligence without requiring expensive tractors or complex sensors.

Feasibility & Scalability

- **Technical Feasibility:** Relies on established IoT telemetry protocols (MQTT/HTTP) combined with standard agronomic regression models.
- **Scalability:** Cloud-native microservices allow rapid deployment across varied agro-climatic zones and diverse crop types.
- **Risks & Mitigations:** Addressing rural connectivity limits through SMS fallbacks and cached offline algorithmic schedules.

Cost for Product

- **Edge Hardware (Optional Sensor Node):** 1200 INR (ESP32 micro-controller, capacitive soil moisture probe, solar lipo charger).
- **Cloud Infrastructure & API Ingestion:** 1000 INR/month (scalable AWS/GCP tier, weather data caching).
- **End-User Delivery Cost:** Freely accessible web app, optional low-cost 100 INR/season SMS alert subscription.

Use Cases

- **Use Case 1: Runoff Avoidance:** Smallholder paddy farmer receives a 48-hour unseasonal rain alert advising to delay nitrogen top-dressing, saving money
- **Use Case 2: Dry-Soil Burn Prevention:** Maize grower receives an alert that soil moisture is below the 20% threshold, preventing crop root scorch until after planned irrigation.
- **Use Case 3: Cooperative Planning:** Farmer Producer Organization (FPO) aggregates village-level optimum application windows to bulk-distribute fertilizer right in time.

Impacts and Benefits

- **Target Farmer Benefit:** 20–30% reduction in aggregate fertilizer expenditure while preserving expected harvest volume.
- **Environmental Impact:** Substantial decline in groundwater nitrate contamination and nitrous oxide (N_2O) emissions.
- **Measurable Outcome:** 15–20% lift in Nutrient Use Efficiency (NUE) within the first cultivation cycle.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 2	Target 2.4	Zero Hunger	Promotes sustainable agricultural practices that increase productivity and resilience.
SDG 12	Target 12.4	Responsible Consumption	Minimizes excessive chemical runoff into soil, waterways, and air.

Individual Member Contribution

Member	Contribution or responsibility
Austin Binoy	Agronomic algorithm logic, climate modeling, and data pipelines
Siddharth HariPanicker	IoT sensor hardware prototyping and soil telemetry calibration
Ashwin Shajil	Backend architecture (FastAPI), database schemas, and API integration
H Dheeraj	UI/UX dashboard design, mobile client development, and testing

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